References

[1] Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. 2023. Self-Consuming Generative Models Go MAD. https://doi.org/10.48550/arXiv.2307.01850 arXiv:2307.01850 [cs].

[2] Danial Alihosseini, Ehsan Montahaei, and Mahdieh Soleymani Baghshah. 2019. Jointly Measuring Diversity and Quality in Text Generation Models. In *Proceedings of the Workshop on Methods for Optimizing and Evaluating Neural Language Generation*, Antoine Bosselut, Asli Celikyilmaz, Marjan Ghazvininejad, Srinivasan Iyer, Urvashi Khandelwal, Hannah Rashkin, and Thomas Wolf (Eds.). Association for Computational Linguistics, Minneapolis, Minnesota, 90–98. https://doi.org/10.18653/v1/W19-2311

[3] Kareem Amin, Sara Babakniya, Alex Bie, Weiwei Kong, Umar Syed, and Sergei Vassilvitskii. 2025. Escaping Collapse: The Strength of Weak Data for Large Language Model Training. https://doi.org/10.48550/arXiv.2502.08924 arXiv:2502.08924 [cs].

[4] Daniela Anderson, Katie Lien, Chibueze Agwu, Phillip S. Ang, and Nabil Abou Baker. 2023. The Bias of Medicine in Sickle Cell Disease. *Journal of General Internal Medicine* 38, 14 (Nov. 2023), 3247–3251. https://doi.org/10.1007/s11606-023-08392-0

[5] Richard Antonello, Nicole Beckage, Javier Turek, and Alexander Huth. 2021. Selecting Informative Contexts Improves Language Model Fine-tuning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics, Online, 1072–1085. https://doi.org/10.18653/v1/2021.acl-long.87

[6] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. 2022. Constitutional AI: Harmlessness from AI Feedback. http://arxiv.org/abs/2212.08073 arXiv:2212.08073 [cs].

[7] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*. ACM, Virtual Event Canada, 610–623. https://doi.org/10.1145/3442188.3445922

[8] Quentin Bertrand, Avishek Joey Bose, Alexandre Duplessis, Marco Jiralerspong, and Gauthier Gidel. 2024. On the Stability of Iterative Retraining of Generative Models on their own Data. https://doi.org/10.48550/arXiv.2310.00429 arXiv:2310.00429 [cs].

[9] Borhane Blili-Hamelin, Christopher Graziul, Leif Hancox-Li, Hananel Hazan, El-Mahdi El-Mhamdi, Avijit Ghosh, Katherine A Heller, Jacob Metcalf, Fabricio Murai, Eryk Salvaggio, Andrew J Smart, Todd Snider, Mariame Tighanimine, Talia Ringer, Margaret Mitchell, and Shiri Dori-Hacohen. 2025. Position: Stop treating 'AGI' as the north-star goal of AI research. In *Forty-second International Conference on Machine Learning Position Paper Track*. https://openreview.net/forum?id=1RlrtH6ydW

[10] Rishi Bommasani, Kathleen A. Creel, Ananya Kumar, Dan Jurafsky, and Percy S. Liang. 2022. Picking on the same person: Does algorithmic monoculture lead to outcome homogenization? *Advances in Neural Information Processing Systems* 35 (2022), 3663–3678. https://proceedings.neurips.cc/paper_files/paper/2022/hash/17a234c91f746d9625a75cf8a8731ee2-Abstract-Conference.html

[11] Gavin Brown, Jeremy Wyatt, Rachel Harris, and Xin Yao. 2005. Diversity creation methods: a survey and categorisation. *Information fusion* 6, 1 (2005), 5–20. https://www.sciencedirect.com/science/article/pii/S1566253504000375 Publisher: Elsevier.

[12] John Chung, Ece Kamar, and Saleema Amershi. 2023. Increasing Diversity While Maintaining Accuracy: Text Data Generation with Large Language Models and Human Interventions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (Eds.). Association for Computational Linguistics, Toronto, Canada, 575–593. https://doi.org/10.18653/v1/2023.acl-long.34

[13] Sasha Costanza-Chock. 2020. *Design justice: Community-led practices to build the worlds we need.* The MIT Press. https://library.oapen.org/handle/20.500.12657/43542

[14] Rohan Deb, Kiran Thekumparampil, Kousha Kalantari, Gaurush Hiranandani, Shoham Sabach, and Branislav Kveton. 2025. FisherSFT: Data-Efficient Supervised Fine-Tuning of Language Models Using Information Gain. https://doi.org/10.48550/arXiv.2505.14826 arXiv:2505.14826 [cs].

[15] Thomas G. Dietterich. 2000. Ensemble Methods in Machine Learning. In *Multiple Classifier Systems*, Gerhard Goos, Juris Hartmanis, and Jan Van Leeuwen (Eds.). Vol. 1857. Springer Berlin Heidelberg, Berlin, Heidelberg, 1–15. https://doi.org/10.1007/3-540-45014-9_1 Series Title: Lecture Notes in Computer Science.

[16] Elvis Dohmatob, Yunzhen Feng, Arjun Subramonian, and Julia Kempe. 2024. Strong Model Collapse. https://doi.org/10.48550/arXiv.2410.04840 arXiv:2410.04840 [cs].

[17] Elvis Dohmatob, Yunzhen Feng, Pu Yang, Francois Charton, and Julia Kempe. 2024. A Tale of Tails: Model Collapse as a Change of Scaling Laws. https://doi.org/10.48550/arXiv.2402.07043 arXiv:2402.07043 [cs].

[18] Heather E. Douglas. 2009. *Science, Policy, and the Value-Free Ideal.* University of Pittsburgh Pre. Google-Books-ID: LcFvKeOJRmgC.

[19] William Falcon and The PyTorch Lightning team. 2019. *PyTorch Lightning.* https://doi.org/10.5281/zenodo.3828935

[20] Sina Fazelpour and Daniel Steel. 2022. Diversity, Trust, and Conformity: A Simulation Study. *Philosophy of Science* 89, 2 (April 2022), 209–231. https://doi.org/10.1017/psa.2021.25 Publisher: Cambridge University Press (CUP).

[21] Shangbin Feng, Taylor Sorensen, Yuhan Liu, Jillian Fisher, Chan Young Park, Yejin Choi, and Yulia Tsvetkov. 2024. Modular Pluralism: Pluralistic Alignment via Multi-LLM Collaboration. http://arxiv.org/abs/2406.15951 arXiv:2406.15951 [cs].

[22] Yunzhen Feng, Elvis Dohmatob, Pu Yang, Francois Charton, and Julia Kempe. 2024. Beyond Model Collapse: Scaling Up with Synthesized Data Requires Verification. https://doi.org/10.48550/arXiv.2406.07515 arXiv:2406.07515 [cs].

[23] Jillian Fisher, Ruth E. Appel, Chan Young Park, Yujin Potter, Liwei Jiang, Taylor Sorensen, Shangbin Feng, Yulia Tsvetkov, Margaret E. Roberts, Jennifer Pan, Dawn Song, and Yejin Choi. 2025. Political Neutrality in AI Is Impossible- But Here Is How to Approximate It. https://doi.org/10.48550/arXiv.2503.05728 arXiv:2503.05728 [cs].

[24] Miranda Fricker. 2007. *Epistemic Injustice: Power and the Ethics of Knowing.* Clarendon Press. Google-Books-ID: lncSDAAAQBAJ.

[25] Shi Fu, Yingjie Wang, Yuzhu Chen, Xinmei Tian, and Dacheng Tao. 2025. A Theoretical Perspective: How to Prevent Model Collapse in Self-consuming Training Loops. https://doi.org/10.48550/arXiv.2502.18865 arXiv:2502.18865 [cs].

[26] M. A. Ganaie, Minghui Hu, A. K. Malik, M. Tanveer, and P. N. Suganthan. 2022. Ensemble deep learning: A review. *Engineering Applications of Artificial Intelligence* 115 (Oct. 2022), 105151. https://doi.org/10.1016/j.engappai.2022.105151 arXiv:2104.02395 [cs].

[27] Timnit Gebru and Émile P. Torres. 2024. The TESCREAL bundle: Eugenics and the promise of utopia through artificial general intelligence. *First Monday* (April 2024). https://doi.org/10.5210/fm.v29i4.13636

[28] Matthias Gerstgrasser, Rylan Schaeffer, Apratim Dey, Rafael Rafailov, Henry Sleight, John Hughes, Tomasz Korbak, Rajashree Agrawal, Dhruv Pai, Andrey Gromov, Daniel A. Roberts, Diyi Yang, David L. Donoho, and Sanmi Koyejo. 2024. Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data. https://doi.org/10.48550/arXiv.2404.01413 arXiv:2404.01413 [cs].

[29] Mandeep Goyal and Qusay H. Mahmoud. 2024. A Systematic Review of Synthetic Data Generation Techniques Using Generative AI. *Electronics* 13, 17 (Jan. 2024), 3509. https://doi.org/10.3390/electronics13173509 Publisher: Multidisciplinary Digital Publishing Institute.

[30] Qianyue Hao, Fengli Xu, Yong Li, and James Evans. 2026. Artificial intelligence tools expand scientists' impact but contract science's focus. *Nature* 649, 8099 (Jan. 2026), 1237–1243. https://doi.org/10.1038/s41586-025-09922-y

[31] David Herel and Tomas Mikolov. 2024. Collapse of Self-trained Language Models. http://arxiv.org/abs/2404.02305 arXiv:2404.02305 [cs].

[32] M. O. Hill. 1973. Diversity and Evenness: A Unifying Notation and Its Consequences. *Ecology* 54, 2 (1973), 427–432. https://doi.org/10.2307/1934352 _eprint: https://esajournals.onlinelibrary.wiley.com/doi/pdf/10.2307/1934352.

[33] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. 2022. Training Compute-Optimal Large Language Models. https://doi.org/10.48550/arXiv.2203.15556 arXiv:2203.15556 [cs].

[34] Shomik Jain, Vinith Suriyakumar, Kathleen Creel, and Ashia Wilson. 2024. Algorithmic Pluralism: A Structural Approach To Equal Opportunity. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT '24)*. Association for Computing Machinery, New York, NY, USA, 197–206. https://doi.org/10.1145/3630106.3658899

[35] Jiaming Ji, Boyuan Chen, Hantao Lou, Donghai Hong, Borong Zhang, Xuehai Pan, Tianyi Alex Qiu, Juntao Dai, and Yaodong Yang. 2024. Aligner: Efficient alignment by learning to correct. *Advances in Neural Information Processing Systems* 37 (2024), 90853–90890. https://proceedings.neurips.cc/paper_files/paper/2024/hash/a51a74b2d71387dc71cc29181b5519bb-Abstract-Conference.html

[36] Lou Jost. 2006. Entropy and diversity. *Oikos* 113, 2 (2006), 363–375. https://doi.org/10.1111/j.2006.0030-1299.14714.x _eprint: https://nsojournals.onlinelibrary.wiley.com/doi/pdf/10.1111/j.2006.0030-1299.14714.x.

[37] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling Laws for Neural Language Models. https://doi.org/10.48550/arXiv.2001.08361 arXiv:2001.08361 [cs].

[38] Jon Kleinberg and Manish Raghavan. 2021. Algorithmic monoculture and social welfare. *Proceedings of the National Academy of Sciences* 118, 22 (June 2021), e2018340118. https://doi.org/10.1073/pnas.2018340118

[39] Anders Krogh and Jesper Vedelsby. 1994. Neural network ensembles, cross validation, and active learning. *Advances in neural information processing systems* 7 (1994). https://proceedings.neurips.cc/paper/1994/hash/b8c37e33defde51cf91e1e03e51657da-Abstract.html

[40] Seth Lazar and Alondra Nelson. 2023. AI safety on whose terms? *Science* 381, 6654 (July 2023), 138–138. https://doi.org/10.1126/science.adi8982

[41] Weixin Liang, Yaohui Zhang, Zhengxuan Wu, Haley Lepp, Wenlong Ji, Xuandong Zhao, Hancheng Cao, Sheng Liu, Siyu He, Zhi Huang, Diyi Yang, Christopher Potts, Christopher D. Manning, and James Zou. 2025. Quantifying large language model usage in scientific papers. *Nature Human Behaviour* (Aug. 2025), 1–11. https://doi.org/10.1038/s41562-025-02273-8 Publisher: Nature Publishing Group.

[42] Sijia Liu, Yuanshun Yao, Jinghan Jia, Stephen Casper, Nathalie Baracaldo, Peter Hase, Yuguang Yao, Chris Yuhao Liu, Xiaojun Xu, Hang Li, Kush R. Varshney, Mohit Bansal, Sanmi Koyejo, and Yang Liu. 2025. Rethinking machine unlearning for large language models. *Nature Machine Intelligence* 7, 2 (Feb. 2025), 181–194. https://doi.org/10.1038/s42256-025-00985-0

[43] Helen E Longino. 1990. *Science as social knowledge: Values and objectivity in scientific inquiry*. Princeton university press.

[44] Christina Lu and Max Van Kleek. 2024. Model Plurality: A Taxonomy for Pluralistic AI. (Oct. 2024). https://openreview.net/forum?id=kil2mabTqx

[45] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2016. Pointer sentinel mixture models. *arXiv preprint arXiv:1609.07843* (2016).

[46] Lisa Messeri and M. J. Crockett. 2024. Artificial intelligence and illusions of understanding in scientific research. *Nature* 627, 8002 (March 2024), 49–58. https://doi.org/10.1038/s41586-024-07146-0 Publisher: Nature Publishing Group.

[47] Mozilla. 2024. Mozilla Report: How Common Crawl's Data Infrastructure Shaped the Battle Royale over Generative AI. https://www.mozillafoundation.org/en/blog/Mozilla-Report-How-Common-Crawl-Data-Infrastructure-Shaped-the-Battle-Royale-over-Generative-AI/ Section: AI.

[48] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. (2022).

[49] Andrew J. Peterson. 2025. AI and the problem of knowledge collapse. *AI & SOCIETY* 40, 5 (June 2025), 3249–3269. https://doi.org/10.1007/s00146-024-02173-x

[50] Samuli Pöyhönen. 2017. Value of cognitive diversity in science. *Synthese* 194, 11 (Nov. 2017), 4519–4540. https://doi.org/10.1007/s11229-016-1147-4

[51] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language Models are Unsupervised Multitask Learners. (2019).

[52] Kristina Rolin. 2019. The Epistemic Significance of Diversity. In *The Routledge Handbook of Social Epistemology* (1 ed.), Miranda Fricker, Peter J. Graham, David Henderson, and Nikolaj J. L. L. Pedersen (Eds.). Routledge, 158–166. https://doi.org/10.4324/9781315717937-16

[53] Michael Roswell, Jonathan Dushoff, and Rachael Winfree. 2021. A conceptual guide to measuring species diversity. *Oikos* 130, 3 (2021), 321–338. https://doi.org/10.1111/oik.07202 _eprint: https://nsojournals.onlinelibrary.wiley.com/doi/pdf/10.1111/oik.07202.

[54] Rylan Schaeffer, Joshua Kazdan, Alvan Caleb Arulandu, and Sanmi Koyejo. 2025. Position: Model Collapse Does Not Mean What You Think. https://doi.org/10.48550/arXiv.2503.03150 arXiv:2503.03150 [cs].

[55] Max Schaffelder and Albert Gatt. 2025. Synthetic Eggs in Many Baskets: The Impact of Synthetic Data Diversity on LLM Fine-Tuning. https://doi.org/10.48550/arXiv.2511.01490 arXiv:2511.01490 [cs].

[56] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. AI models collapse when trained on recursively generated data. *Nature* 631, 8022 (July 2024), 755–759. https://doi.org/10.1038/s41586-024-07566-y

[57] Miriam Solomon. 2006. Norms of Epistemic Diversity. *Episteme* 3, 1-2 (June 2006), 23–36. https://doi.org/10.3366/epi.2006.3.1-2.23

[58] Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari S. Morcos. 2023. Beyond neural scaling laws: beating power law scaling via data pruning. https://doi.org/10.48550/arXiv.2206.14486 arXiv:2206.14486 [cs].